Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 2 – Project Management Plan

– Hanoi, Aug 2026 –

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Change Log

Date	A* M, D	In charge	Change Description
30/9/2026	A	Nguyen Hoang Thang	Add I.Project Overview
30/9/2026	A	Nguyen Hoang Thang	Add II.Management Approach
19/10/2026	M	Nguyen Chi Tung Duong	Standardize Scope, Assumptions (Cloudflare R2, Gmail SMTP, FCM)
20/10/2026	M	Nguyen Chi Tung Duong	Standardize WBS, Capacity Planning, Objectives, Risks
21/10/2026	M	Nguyen Chi Tung Duong	Standardize Iterative Plan, and Tools & Infrastructure to align with Report 1 and Report 3

*A - Added M - Modified D - Deleted

I. Project Overview

1. Scope and Purpose

The primary objective of the EcoLink project is to build an Intelligent Recyclable Waste Classification and Trade Matching System. The project addresses core environmental challenges in Vietnam: a widespread lack of household waste-sorting knowledge, the absence of an efficient channel connecting households with verified local scrap collectors and recycling facilities, and low long-term user retention in daily sorting habits. By combining Computer Vision (AI) image recognition, localized sorting instructions, GPS-based recycler discovery, direct handover coordination, and Green Points gamification incentives, EcoLink transforms routine waste disposal into an engaging, eco-friendly, and financially rewarding habit.

Architected as a responsive web application, EcoLink incorporates several key functional modules within its core scope. The system features AI-driven waste classification capable of identifying 8 to 12 common categories of recyclable waste using a trained YOLOv8 model, supported by a user misclassification feedback and Staff dataset curation loop. Classification results trigger a localized Rule Engine that delivers dynamic, step-by-step cleaning and disposal instructions alongside a searchable Waste Catalog and educational Blog. Furthermore, households can discover nearby verified recyclers through spatial querying using PostgreSQL/PostGIS, submit direct handover requests with preferred collection time slots, negotiate terms via an in-app real-time chatbox, confirm digital handover receipts with scale proof photos, rate recyclers, and submit transaction complaints for Staff arbitration. Payment for buying and selling recyclable waste is settled directly between the User and the Recycler outside the system (cash or personal bank transfer); dynamic VietQR integration is used strictly for purchasing internal platform service packages (such as Ad-Free experience and Verified Partner badges), accompanied by Admin financial reconciliation, RBAC staff governance, and executive analytics.

2. Assumptions & Constraints

EcoLink is an independent capstone project developed by group **SEP490-G141** during the Fall 2026 semester at FPT University. It does not inherit legacy codebases from prior university projects and operates as a standalone software platform.

No	Description	Note
Assumptions		
1	All 5 team members maintain dedicated availability (56 hours/week across 14 weeks, equivalent to 490 gross person-days) to fulfill the expanded 18-feature project scope.	Resources / Capacity
2	Lecturer / Supervisor (Mr. Ta Dinh Tien) reviews and provides feedback on milestone deliverables within 3–5 business days.	Review & Approval
3	Publicly available datasets (Kaggle Garbage 12-Class, TrashNet, TACO) supplemented by 40% self-collected real-world photos provide sufficient training quality for the YOLOv8 AI model.	Technical / AI Data

4	Open-source software components and free-tier external services (.NET 10 SDK, Python FastAPI, PostgreSQL/PostGIS, Docker, Cloudflare R2 Object Storage, Gmail SMTP, Firebase Cloud Messaging, Google OAuth 2.0, OpenStreetMap/Nominatim, and VietQR sandbox) remain stable and accessible.	Technology Stack
Constraints		
1	The final software package, documentation, and oral defense must be completed strictly within the 14-week semester timeline (September 2026 – December 2026).	Schedule
2	Zero hardware/license budget; the system must be built and deployed using free-tier cloud infrastructure (Cloudflare R2, Gmail SMTP, FCM, OSM) and local/VPS Docker containerization.	Budget
3	System core architecture must follow Clean Architecture (.NET) combined with a dedicated Python FastAPI AI Vision Service.	Technical Architecture
4	System must comply with standard web application security rules (RBAC permission matrix, salted password hashing, JWT authentication, and restricted access to private handover receipts and verification documents).	Security

3. Cost & Time Estimations

#	Work Package	Effort (pds)	Due Date
1	**Stage 1: Project Initiating (Week 1)**	**TBD**	**13/09/2026**
1.1	Project Team Setup, Topic Selection & Supervisor Alignment	TBD	11/09/2026
1.2	Report 1: Project Introduction v1.0	TBD	13/09/2026
2	**Stage 2: Project Planning & Initial Requirements (Weeks 2–3)**	**TBD**	**27/09/2026**
2.1	AI Feasibility Survey & System Use Cases Definition (52 UCs / 54 Sub-flows)	TBD	15/09/2026
2.2	User & Recycler Requirements Survey (Google Forms & Field Work)	TBD	24/09/2026
2.3	Report 2: Project Management Plan v1.0	TBD	26/09/2026
2.4	Report 3: Software Requirement Specification (SRS v0.9)	TBD	27/09/2026
3	**Stage 3: Software Design & Proof of Concept (Weeks 4–5)**	**TBD**	**11/10/2026**

3.1	Vertical Prototype / AI Vision Service POC (YOLOv8 + FastAPI)	TBD	04/10/2026
3.2	Report 4: Software Design Specification (SDS v1.0 - Architecture & ERD)	TBD	08/10/2026
3.3	Report 5: Test Planning & Test Strategy	TBD	11/10/2026
3.4	Report 3: SRS v1.0 (Finalized Requirements for Implementation)	TBD	11/10/2026
4	**Stage 4: Implementation & Testing (Weeks 6–11)**	**TBD**	**22/11/2026**
4.1	Iteration 1: Core AI Vision, Rule Engine, Catalog & Auth (Weeks 6–7)	TBD	25/10/2026
4.1.1	Backend Base Architecture (.NET Clean Architecture), JWT/Google OAuth & Gmail SMTP	TBD	18/10/2026
4.1.2	AI Waste Scan Service (Cloudflare R2 + FastAPI) & Rule Engine / Waste Catalog (FE & BE)	TBD	25/10/2026
4.1.3	Unit Testing & Iteration 1 Software Package v1.0	TBD	25/10/2026
4.2	Iteration 2: Recycler Onboarding, GPS Discovery, Handover & Chatbox (Weeks 8–9)	TBD	08/11/2026
4.2.1	Recycler Upgrade Onboarding, Staff Verification & PostGIS Nearby Recycler Discovery	TBD	01/11/2026
4.2.2	Handover Request Coordination, In-App Negotiation Chatbox & Digital Handover Receipts	TBD	08/11/2026
4.2.3	Integration Testing & Iteration 2 Software Package v2.0	TBD	08/11/2026
4.3	Iteration 3: Gamification, VietQR Services, Governance & AI Curation (Weeks 10–11)	TBD	22/11/2026
4.3.1	Green Points Check-in, Virtual Tree, Eco Facts, Gift Store & Leaderboard Cycles	TBD	15/11/2026
4.3.2	Ratings, Complaint Arbitration, In-System VietQR Service Packages, Blog & AI Dataset Curation	TBD	22/11/2026
4.3.3	Admin Analytics Dashboard, Report Export, Staff RBAC & Software Package v3.0	TBD	22/11/2026
5	**Stage 5: Verification & Validation (Weeks 12–13)**	**TBD**	**06/12/2026**
5.1	System Testing, Performance Testing & Bug Fixing	TBD	29/11/2026

5.2	Report 6: User Guides & Installation Manuals	TBD	03/12/2026
5.3	Final Software Package Freeze (Documents & Source Code Deployment)	TBD	06/12/2026
6	**Stage 6: Project Closing (Week 14)**	**10**	**13/12/2026**
6.1	Report 7: Final Project Report	5	10/12/2026
6.2	Slide Presentation & Final Capstone Defense Preparation	5	13/12/2026
TOTAL	**Entire Project Lifecycle (14 Weeks)**	**450 pds**	**13/12/2026**

4. Capacity Planning

Item	Value
Hours per person per week	56 h (8 h/day × 7 days)
Team total per week	280 h/week (5 people × 56 h/week)
Total duration	14 weeks
Gross capacity	3920 h (490 person-days)
Planned project effort	3600 h (450 person-days)
Contingency & review buffer (~8.2%)	320 h (40 person-days)

5. Project Objectives

#	Metric	Unit	Planned	Actual	Notes / References
1	**Total Effort Usage**	Person-Day (pds)	≤ 450 pds	TBD	Measured against gross capacity of 490 pds
2	**Requirement Completeness**	Percentage (%)	≥ 95%	TBD	Ratio of implemented use cases to approved SRS (52 UCs; Fail if < 75%)
3	**Schedule Timeliness**	Percentage (%)	100%	TBD	On-time deliverables / Total planned milestone deliverables

4	**AI Model Accuracy (Top-1)**	Percentage (%)	≥ 85%	TBD	Tested on 8–12 common recyclable waste categories
5	**AI Low-Confidence Threshold**	Confidence Score	Admin-configurable (default < 60%)	TBD	Flags uncertain predictions and prompts manual category confirmation (BR-CLS-03)
6	**Web Page Load Time**	Seconds (s)	≤ 3.0 s	TBD	95th percentile non-media page load under up to 100 concurrent active users (NFR-PRF)
7	**Review Defects (Severity 1 & 2)**	Number of defects	0 open	TBD	Critical defects found during supervisor reviews must be resolved
8	**System Test Defects Resolution**	Percentage (%)	100% High/Critical	TBD	All blocker defects must be fixed before final defense

6. Risks and Mitigations

#	Risk	Prob	Impact	Mitigation
R01	**Low AI Accuracy for Ambiguous / Damaged Waste** AI fails to correctly distinguish crumpled paper, dirty plastic, or complex items.	Medium	High	**1. Restrict initial model scope to 8–12 well-defined recyclable classes.** **2. Enforce an Admin-configurable confidence threshold (default 60%) requiring user confirmation or manual selection.** **3. Provide a misclassification report & Staff dataset curation workflow (FE-15) to continuously retrain the YOLOv8 model.**
R02	**AI Inference Latency / Timeout during Live Demo** FastAPI AI service experiences high latency or timeout (> 15s) during defense.	Medium	High	**1. Deploy the FastAPI YOLOv8 service in a local/same-VPC Docker container with pre-warmed model weights.** **2. Enforce a 15-second hard timeout with graceful fallback to manual Waste Catalog search (U-03) and pre-tested demo image sets.**

R03	**Scope Creep & Payment Boundary Confusion** Uncontrolled additions of complex features (e.g., in-system trade payout escrow/disbursement or native mobile apps).	Medium	High	**1. Formally freeze SRS after Week 5 review.** **2. Explicitly define Exclusions (LI-01 to LI-10) in Report 1: waste trade settlement happens directly outside the system; VietQR is strictly limited to internal service packages.** **3. Require impact analysis and supervisor approval for any Change Request.**
R04	**PostGIS Spatial Discovery Performance Bottleneck** Slow response when querying nearby recyclers by GPS coordinates under concurrent load.	Low	Medium	**1. Apply GiST spatial indexing on PostGIS geography columns.** **2. Bound search radius to 2–20 km and filter only approved, Online recyclers prior to distance sorting.**
R05	**Insufficient Real-World User & Recycler Survey Data** Difficulty reaching enough local scrap collectors (vựa ve chai) for survey data.	Medium	Medium	**1. Assign a dedicated 3-member sub-team (Luan, Duong, Hoang).** **2. Conduct direct field interviews at local scrap yards in Hanoi.** **3. Utilize online Google Forms targeting eco-conscious communities.**

R06	**Third-Party Service Disruption / Rate Limits** Temporary outage or quota exhaustion on Gmail SMTP, VietQR webhook, OpenStreetMap Nominatim, or FCM.	Low	Medium	**1. Implement asynchronous retry queues (up to N retries) for Gmail SMTP and FCM so transient failures never roll back core business transactions (BR-NTF-06).** **2. Use sandbox/mock webhook verification for VietQR testing and cache geocoded addresses.**
R07	**High Workload Intensity (450 pds across 18 Features)** Tight buffer (~8.2%) against gross team capacity over the 14-week timeline.	Medium	High	**1. Prioritize core AI scan, Rule Engine, GPS discovery, and Handover Request workflows in Iterations 1–2.** **2. Track weekly burn-down closely in Daily Sync meetings and reallocate cross-functional members early if bottlenecks arise.**

II. Management Approach

1. Project Process/Methodology

The EcoLink project applies a **Hybrid Software Development Methodology**, combining Waterfall for project initiation, requirement freezing, and design phases with Agile/Scrum Iterative Development for software implementation. This approach ensures structured compliance with academic milestones while maintaining flexibility during feature implementation.

Initiation & Requirement Planning Phase (Weeks 1–3):

- Establish core team roles, set up Git repositories, and align with Supervisor (Mr. Ta Dinh Tien).
- Conduct real-world user surveys, perform gap analysis against existing market solutions, and formalize the Software Requirement Specification (SRS v0.9/v1.0 with 52 Use Cases) and Project Management Plan (PMP).

Architecture & Detailed Design Phase (Weeks 4–5):

1. Build a Vertical Prototype / Proof of Concept (POC) for the Python FastAPI YOLOv8 AI Vision Service.
2. Construct System Design Specifications (SDS v1.0), ERD schemas, API inventories, and Test Plans.

Iterative Implementation Phase (Weeks 6–11):

1. **Iteration 1 (Weeks 6–7):** Core Backend setup (.NET Clean Architecture), JWT & Google OAuth 2.0 Authentication, Gmail SMTP verification, AI Waste Classification API (Cloudflare R2 + FastAPI YOLOv8), Waste Catalog search, and Rule Engine sorting guidance (FE-01, FE-02, FE-03, FE-12).
2. **Iteration 2 (Weeks 8–9):** Recycler upgrade onboarding & Staff verification, Recycler profile/pricing/online toggle, PostGIS GPS nearby recycler discovery, Handover Request coordination, In-App Negotiation Chatbox, Digital Handover Receipts (external trade settlement), and Multi-channel Notifications (FE-04 to FE-09, FE-11).
3. **Iteration 3 (Weeks 10–11):** Partner Rating & Complaint arbitration, Green Points Gamification (Daily Check-in, Virtual Tree, Eco Facts, Gift Store, Leaderboard cycles), In-System VietQR Service Packages & financial reconciliation, AI Misclassification feedback & dataset curation, Blog management, Staff RBAC governance, and Admin Analytics Dashboard & Report Export (FE-10, FE-13 to FE-18).

Verification, Validation & Freeze Phase (Weeks 12–13):

1. Execute System Testing, Integration Testing, Performance/Latency verification, and User Manual writing.
2. Final freeze of source code and deployment artifacts.

Project Closing Phase (Week 14):

1. Finalize Capstone Project Report (Report 7) and conduct final presentation defense.

2. Quality Management

To achieve the project metrics defined in Section I (such as ≥ 85% AI top-1 classification accuracy, ≤ 3.0s web page load time under 100 concurrent active users, and zero high-severity defects), the team enforces quality controls across two core dimensions:

2.1 Defect Prevention & Code Standards

1. **Clean Architecture Compliance:** Enforce strict separation of layers in the .NET solution (Domain, Application, Infrastructure, WebApi) so that domain logic remains independent of external frameworks.
2. **Code Reviews:** Every Pull Request (PR) merged into the main branch requires at least one code review and approval by the Tech Lead (Thang) or Sub-leader (Giang).
3. **Static Analysis:** Apply standard linter tools (Roslyn analyzers for C# .NET and PEP8 / Flake8 for Python FastAPI).

2.2 Testing Levels & Validation

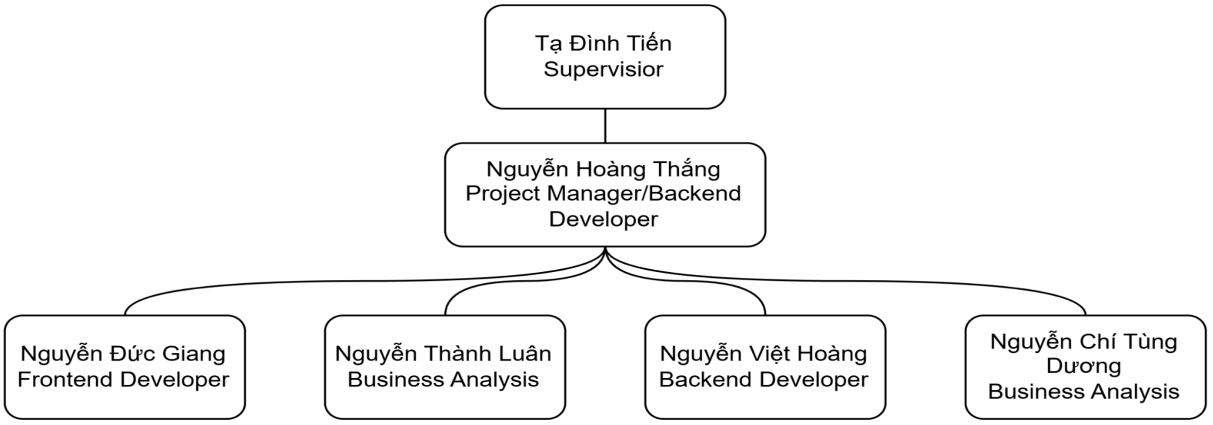
1. **Unit Testing:** Write unit tests for business handlers, CQRS services, and rule mapping calculations using xUnit (.NET).
2. **Integration Testing:** Test API endpoints, EF Core database mappings, PostGIS spatial queries, VietQR IPN webhook idempotency, and inter-service communication between .NET Web API and Python FastAPI.
3. **AI Model & Feedback Loop Validation:** Evaluate YOLOv8 accuracy on a dedicated test dataset using Precision, Recall, F1-Score, and Confusion Matrix analysis. Enforce an Admin-configurable low-confidence threshold (default < 60%) to require manual user confirmation (BR-CLS-03) and route user-reported misclassifications into the Staff dataset curation queue.

3. Project Training Plan

To bridge skill gaps and ensure all team members can execute their assigned work packages efficiently, the following training activities are planned during the initial weeks of the project:

Training Area	Participants	When, Duration
.NET 10 SDK & Clean Architecture	Thang (Leader), Giang, Luan, Hoang, Duong	Week 1 – Week 2 (10 hours)
Python FastAPI & YOLOv8 Computer Vision	Thang (Leader)	Week 2 – Week 3 (12 hours)
PostgreSQL & PostGIS Spatial Querying	Thang, Giang	Week 2 – Week 3 (8 hours)
Git / GitHub Workflow & Docker Compose	Entire Team (5 members)	Week 1 (4 hours)
Figma UI/UX & Responsive Web Design	Giang (Sub-leader)	Week 2 (6 hours)

4. Responsibility Assignments



RACI Chart: R~Responsible, A~Accountable, C~Consulted, I~Informed

Responsibility	THANGNH	GIANGND	LUANNT	HOANGNV	DUONGNCT
Project Planning & Tracking	R	A	I	I	I
Prepare Project Introduction Document	C	A	R	R	R
Prepare SRS Document (Overview Part)	A	C	R	R	R
Prepare SRS Document (User Requirements)	A	C	R	R	R
System Architecture Design	R	A	I	C	I
Database Design	R	A	C	C	C

Backend Development	R	A	C	R	C
Frontend Development	A	R	I	I	I
Integration & API Development	R	A	I	R	I
Test Case Preparation	R	A	I	I	R
Unit Testing	R	A	I	R	I
Integration Testing	A	R	I	R	I
System Testing	A	R	R	R	I
Deployment & Maintenance	R	A	I	I	I

5. Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Daily Sync Meeting	Entire Team (5 members)	Track daily progress, identify blockers, and align daily task assignments.	Mon – Fri (60 mins, 21:30)	Online (Messenger Group / Zalo / Discord)
Weekly Progress Meeting	Entire Team (5 members)	Review weekly deliverables, update Weekly Status Reports, and plan tasks for the next week.	Weekly (Thu 08:00 / Fri 21:00)	Hybrid (Offline Room DE418 / Google Meet)
Supervisor Progress Review	Team & Supervisor (Mr. Ta Dinh Tien)	Report milestone progress, demo prototypes, submit report drafts, and receive feedback.	Bi-weekly / Milestone checkpoints	Offline Meeting (Room DE418, FPT University)

6. Configuration Management

6.1 Document Management

Centralized Storage: All project documents (proposal, SRS, SDS, weekly reports, user guides) are stored and collaborated on via Google Drive (SEP490_G141_FA26).

File Naming Convention: Documents follow the strict naming structure: [Document_Name]_[Group_ID]_[Version]_[Date]

Example: Report1_SEP490_G141_v1.0.docx, Report2_Project_Tracking_SEP490_G141.xlsx.

Version Control Rules:

- v0.x: Draft versions created during team working sessions.
- v1.0, v2.0: Finalized versions approved by the Leader (Thang) and submitted to the Lecturer.

Audit Trail: Any modification to milestone documents must be recorded in the **Record of Changes / Change Log** table at the beginning of each report.

6.2 Source Code Management

Version Control System: Source code is hosted on GitHub across two dedicated repositories:

1. Backend & AI Service: VPGreenLight/intelligent-waste-sorting-trading
2. Frontend Web: VPGreenLight/ecolink-web

Branching Strategy (Git Flow):

1. main: Production-ready code; protected branch requiring approval from Tech Lead (Thang) or Sub-leader (Giang).
2. develop: Integration branch where features are combined and tested.
3. feature/[feature-id]: Working branch for specific features/use cases (e.g., feature/FE-01-ai-scan, feature/FE-06-postgis-discovery, feature/FE-07-handover-request).
4. bugfix/[bug-id]: Branch for resolving issues identified during testing.

Pull Request (PR) & Commit Rules:

1. Commit messages must follow the format: [FE-xx/UC-xx] <type>: <description> (e.g., [FE-01/U-01] feat: integrate YOLOv8 classification endpoint).
2. Every Pull Request requires at least one code review and approval from Leader/Sub-leader before merging into develop or main.

6.3 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	React.js & Tailwind CSS (Frontend Web), ASP.NET Core 10.0 (Backend Clean Architecture), Python FastAPI + YOLOv8 (Dedicated AI Vision Service)
Database & Storage	PostgreSQL 16 + PostGIS Extension (Spatial GPS Queries), Cloudflare R2 Cloud Object Storage (S3-compatible API)
External Services	Google OAuth 2.0 (SSO), Gmail SMTP (Transactional Email), Firebase Cloud Messaging - FCM (Web Push), VietQR API (In-System Service Packages), OpenStreetMap & Nominatim (Map Tiles & Geocoding)
IDEs / Editors	Visual Studio 2022, Visual Studio Code, PyCharm
Diagramming	Draw.io, StarUML
Documentation	Google Docs, Google Sheets, Google Slides, MS Office
Version Control	GitHub (Source Code), Google Drive (Documents & Reports)

Deployment Server	Docker Compose (Localhost & Cloud VPS Containerization)
Project Management	Google Sheets (Project Tracking & Weekly Reports), GitHub Projects, ClickUp